

Predicting Ski Jumps using State-Space Model

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Outline

- Introduction
- Modeling Framework and Dataset
 - State-space models and Least squares method
 - Data processing
- Methodology
 - Different approaches and error function
- Main Results
 - Animation app
- Discussion and Conclusion

Motivation

- ▶ Planning of ski jump enlargement and anticipating risky conditions
- ▶ Goal: Predict ski jumps based on external factors



Figure: Flying hill in Planica

- ▶ State-space models (SSM)
- ▶ Least squares method (LS)

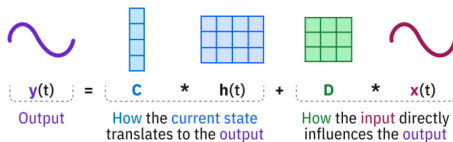
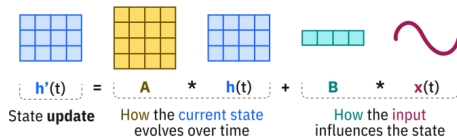


Figure: Schema of SSM

Data Description

- ▶ Cleaning the data
- ▶ Data structure
- ▶ Angles: opening, stalling, yaw, roll
- ▶ wind features : Speed, Tangent, Cross, Turbulence
- ▶ coordinate system on a main ski jumping hill in Planica

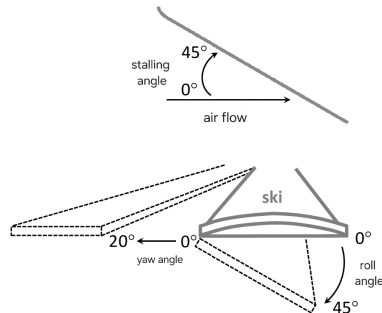


Figure: Representation of different measured angles.

Ski jumping hill

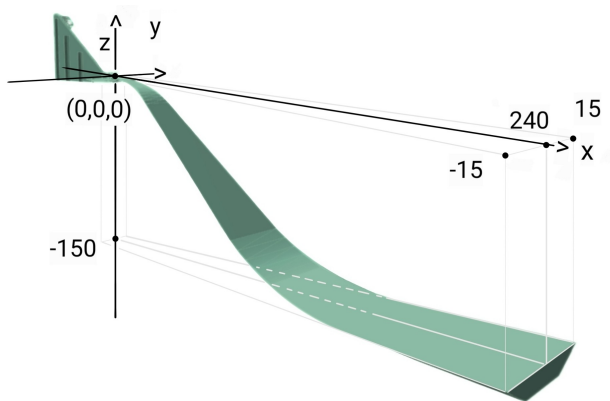


Figure: Ski jumping hill with a coordinate system placed on top.

Different approaches

Different approaches:

- ▶ SSM + PINN (Physics Informed Neural Networks)
- ▶ pure SSM:
 - ▶ State vector (x, y, z coordinates and their speed, jumper's speed, and angles)
 - ▶ Observation vector (x, y, z , height)
 - ▶ Controls vector (averaged wind features and angles)

Average file starter:

- ▶ interpolation over the data points
- ▶ average number of time stamps
- ▶ first realistic input of the jump

Error function

Error function:

- ▶ Error function
- ▶ leave-one-out cross-validation

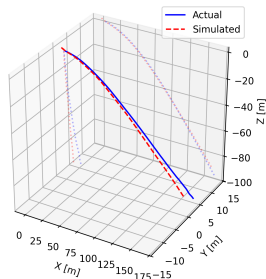
different add-ons to the model:

- ▶ all 12 wind sensors or only 3 wind zones
- ▶ squared wind values

Flight simulation example

Model	Training Error	Test Error
SSM + wind zones	1.32 m	1.37 m
SSM + wind zones + average file	1.58 m	1.63 m
SSM	1.62 m	1.82 m
SSM + squared wind features	1.33 m	1.43 m

Actual vs Simulated Trajectory
Error: 1.361, Simulated length: 185.0, Actual length: 192.1



Application

- Description of the application
- Observing measured and predicted jumps

Ski jump animation

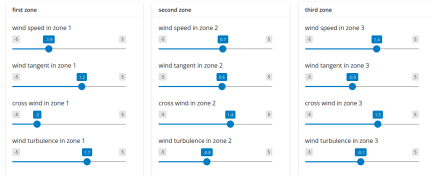
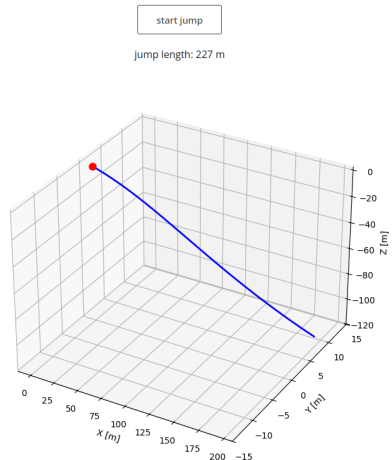


Figure: Adjusted sliders in the app



Interpretation

- ▶ Limitations
- ▶ Future work and potential improvements

Thank you!